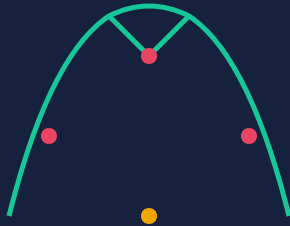


Elliptic Curves and Modular Forms

Elliptic Curve



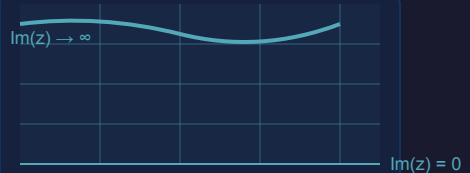
$$y^2 = x^3 - x$$

Group Law: $P + Q + R = 0$

Modularity

$$L(E, s) = L(f, s)$$

Modular Form



$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z)$$

$$q\text{-expansion: } f(z) = \sum a_n q^n$$

Wiles's Theorem (1995)

Every semistable elliptic curve over $\mathbb{Q}$ is modular.

This implies Fermat's Last Theorem (via Ribet's Theorem).

Proof: Galois representations $\rightarrow$ deformation theory $\rightarrow R = T$

The Taniyama-Shimura-Weil conjecture was proved for semistable curves.